## [Diagnostic value of hysterosalpingography in examination of Fallopian tubes in infertile women].

PMID: 11534279 | Indexed in PubMed

Hysterosalpingography (HSG) is a radiographic examination of endocervical canals, uterine cavity and Fallopian tube with the use of a radiographic contrast medium [1]. This method is an integral part of gynaecological examination and its value has not been underestimated in the modern gynaecological practice. The goal of the study was to evaluate the reliability of HSG in the diagnosis of Fallopian tube and to compare the obtained results with laparoscopic findings. The study included 140 infertile women. HSG was performed in the first half of the cycle, usually on the ninth day, without anaesthesia. The instruments after Schultze were used; 15 mL of Telebrix-contrast was used. Three radiograms were done. Laparoscopic examination was carried out in general endotracheal anaesthesia. A Storz laparoscope was used. $CO_2$ was used for artificial pneumoperitoneum and indigolipstick for tube passage. The obtained findings were elaborated statistically. Descriptive and analytic models were used. $p < 0.05$ and $p < 0.01$ were considered as a risk factor of statistical significance. An approximate time interval between the two procedures was 5.18 months. Normal findings of HSG examination were noted in 53 women (37.9%); tube occlusion in 67 women (47.9%), and peritubal adhesion with tubal passage in 20 (14.3%) patients. A normal finding was found in 56 women (40.0%), tubal occlusion in 64 women (45.7%), and peritubal adhesion with tubal passage in 20 (14.3%) patients. HSG and laparoscopic findings regarding normal tubes were in agreement in 32 women (22.9%), tubal occlusion in 35 women (25.0%) and peritabal adhesion with tubal passage in 5 (3.6%) patients. The best sensitivity of HSG was observed in detection of proximal tubal occlusion (78%), and the smallest in occlusion with the accompanying adhesion (2%). The best specificity of HSG was noted in the diagnosis of combined occlusions (96%) and the smallest in tubal passage with peritubal adhesion (25%). There were 15% of false negative findings and 17.1% of false positive findings. The time interval from one to the other procedure can be considered as an important factor in laparoscopic confirmation or negative HSG findings. With the continuation of the time interval the conditions are made for the aggravation of old and occurrence of new pathological processes in genital internal female organs. The possible causes of differential diagnosis of tubal occlusion between HSG and laparoscopic examination might be: 1) unequal anaesthesia during HSG and laparoscopic examination; 2) different properties of contrast media; 3) anatomic variations in the width of lumen tubes; 4) erroneous interpretation of the results. The sensitivity of HSG in this study was different in various types of tubal passage. In other studies the sensitivity of HSG was from 65% [10] to 96%[7]. The high specificity was found during detection of combined tubal occlusion (96%). The results of other authors were similar [7, 10]. This is a good contribution to the statement that HSG is a useful test of tubal obstruction. A rather high percent of false positive results of HSG was established in this study. The possible reasons might be tubal spasm and endometrial polyp in the area of the uterine opening of the tubes. On the basis of the obtained results, the following conclusions can be drawn: 1) HSG is a simple method for examination of female sterility; 2) HSG and laparoscopy are the complementary methods in the examination of tubal sterility; 3) HSG is inferior in relation to laparoscopy in the examination of peritubal adhesion.

## Hysterosalpingography findings of female partners of infertile couple attending fertility clinic at Lagos University Teaching Hospital.

PMID: 35145585 | Indexed in PubMed

Hysterosalpingography (HSG) is an outpatient fluoroscopic method for the evaluation of the uterine cavity, fallopian tubes, and the surrounding peritoneal cavity. Female fertility depends greatly on normal female reproductive organs; hence tubal abnormalities may contribute significantly to female infertility. HSG is an invaluable screening tool in the evaluation of women with suspected tubal factor infertility. This study aims to review the HSG findings of women who sought fertility treatment at the Lagos University Teaching Hospital, Lagos (LUTH). this was a retrospective study of the pattern of HSG findings among female partners of infertile couples seeking fertility treatment at the LUTH, over a 2-year period, from January 2018 to December 2019. a total of 266 medical records and HSG results were reviewed and included in the data analysis. The mean age (± standard deviation) was 38.4 (± 0.3) years with a range of 24 to 50 years. Most (80.5%) of the participants have secondary infertility and majority (65.4%) were nulliparous. Tubal pathology was the commonest abnormality detected on HSG in 54.9% of women. About one-third (30.8%) of women had bilateral tubal occlusion on HSG. With regards to the right fallopian tube, 43.2% of the participants had tubal occlusion, which differs from 41.7% on the left fallopian tube. Similarly, 10.2% of the women had hydrosalpinx on the left tube when compared with 9% on the right tube. Age (OR 1.055; 95% CI: 1.006, 1.106, p-value 0.028), and previous salpingectomy [OR 6.151; 95% CI: 1.335, 28.349] and myomectomy [OR 4.6; 95% CI: 1.814, 11.67] were identified as risk factors for tubal pathologies on HSG. tubal abnormalities are common findings on HSG and the identifiable risk factors for tubal pathologies include age, salpingectomy, and myomectomy. HSG remains a vital screening tool in the evaluation of tubal-factor infertility in Nigeria.

## Hysterosalpingographic evaluation of human immunodeficiency virus-infected and uninfected infertile women.

*PMID: 32284890 | Indexed in PubMed*

Hysterosalpingography (HSG) is an outpatient fluoroscopy-guided procedure that evaluates the uterine cavity and fallopian tube patency in infertile women. Its cost-effective use is being challenged with the human immunodeficiency virus (HIV) burden in KwaZulu-Natal, which characteristically affects multiple organs. The aim of this study was to describe the HSG findings in a group of HIV-infected and uninfected infertile women. This was a retrospective study conducted over a 4-year period (2012-2016) in which the HSG images and reports of 178 infertile patients from records of the Radiology Department were re-reviewed for abnormalities of the cervix, uterus and fallopian tubes. Their clinical data and radiological findings were entered into a pre-coded data sheet and analysed. The frequency of HIV infection amongst patients with infertility was found to be 32.6%. Forty-four patients were on antiretroviral therapy at the time of the HSG examination, whereas three had not yet started treatment. From the 178 HSG reports, 109 (61.2%) were abnormal. Tubal pathologies were the most common abnormalities, accounting for 79 of the 109 cases and was higher in HIV-infected women than in HIV-uninfected women ($p = 0.001$). Uterine filling defects were demonstrated in 13 of the 109 cases. There were two cases of cervical abnormalities. The study demonstrated that tubal abnormalities were the most common findings amongst infertile women undergoing HSG and occurred predominantly in HIV-infected patients.

## Feasibility of dynamic MR-hysterosalpingography for the diagnostic work-up of infertile women.

*PMID: 20482222 | Indexed in PubMed*

Tubal disturbances often contribute to infertility. Conventional hysterosalpingography (HSG) is considered as standard in the assessment of the patency of the fallopian tubes, but requires ionizing radiation and is restricted to the imaging of endoluminal structures. To evaluate dynamic magnetic resonance-HSG (dMR-HSG) in the diagnostic work-up in patients with infertility. Thirty-seven consecutive infertile women underwent dMR-HSG: 20 ml of gadolinium-polyvidone solution (18.4 mM Dotarem 1:20 with polyvidone) were injected intracervically through a 5-Charriere balloon catheter while acquiring five consecutive flash-3D T1-weighted MR sequences with fat saturation. Two experienced readers assessed image quality and anatomic-pathologic correlations prospectively. The relevance of results was evaluated in the clinical context of each patient. Patient comfort was evaluated with a standardized questionnaire. dMR-HSG was successfully completed in 33/37 patient with an average study time of 45 min. In 4 of 37 patients the catheter became dislodged during the examination, resulting in two complete diagnostic failures. Failure in another two patients was due to preliminary termination because of excessive pain and discomfort during the application of the contrast solution. The uterine cavity was completely visualized and bilateral fallopian tube patency was confirmed by dMR-HSG in 27 of 33 patients. Bilateral tubal occlusion was diagnosed in one of the remaining six patients and was confirmed by laparoscopy. Successful selective tubal catheterization was performed in one additional patient with unilateral and one patient with bilateral fallopian tube occlusion. In three cases, the catheter became dislocated at the end of the examination without demonstration of tubal patency. Since all three patients refused diagnostic laparoscopy and conventional HSG, possible bilateral occlusions of the fallopian tubes could not be further assessed. dMR-HSG with cervical cannulation and intracavitary gadolinium injection is feasible and allows assessment of the uterus, the fallopian tubes, and extra-uterine pelvic structures, while avoiding all ionizing radiation in infertile women aiming at pregnancy.

## Conventional and magnetic resonance hysterosalpingography in assessing tubal patency-A comparative study.

*PMID: 31367087 | Indexed in PubMed*

Tubal factors, one of the leading causes of female infertility, have been conventionally evaluated by hysterosalpingography (HSG). The role of magnetic resonance imaging (MRI) in assessing female infertility is gaining importance because of its inherent efficiency in detecting structural abnormalities. Magnetic resonance hysterosalpingography (MR HSG) is less invasive and avoids exposure of ovaries to ionizing radiation. Its utility is extrapolated to visualize fallopian tubes. To assess the diagnostic accuracies of dynamic MR HSG and conventional HSG (cHSG) in identifying tubal patency in women with infertility using diagnostic laparoscopy (DL) as gold standard. A prospective study of 40 patients was conducted over a period of 6 months. The patients were subjected to MR HSG followed by cHSG during the preovulatory period. If tubes were blocked, the patients were subjected to DL in the next menstrual cycle. If the tubes were patent and there was failure of conception, they were subjected to DL in the interval of 3 months. Twenty-four patients had bilateral tubal spill which was confirmed using cHSG and DL. One patient had discordant MR HSG and cHSG results and six patients had discordant MR HSG and DL results. No statistical difference was observed between MR HSG and cHSG. Pelvic MRI is an inevitable tool in infertility evaluation. MR HSG can be used in addition as it avoids exposure of the reproductive organs to radiation and has the same efficacy as cHSG.

## Randomized double-blind clinical trial of eutectic mixture of local anesthetic creams in

## reducing pain during hysterosalpingography.

*PMID: 25780541 | Indexed in PubMed*

Hysterosalpingography (HSG) is considered as a primary test in infertility work up worldwide due to its reliability in evaluating abnormalities related to the uterus and fallopian tubes. To assess the efficacy of applying eutectic mixture of local anesthetics (lidocaine-prilocaine cream) (EMLA) on the uterine cervix in reducing pain during HSG. Eighty patients undergoing HSG as part of infertility evaluation were randomly allocated to groups receiving either EMLA (N = 40) or placebo cream (N = 40) in a double-blinded prospective study. Fifteen minutes before HSG, 5 grams of 5% cream was applied to the uterine cervix using a cervical applicator. The degree of pain experienced by the patient was evaluated during and after HSG at five predefined steps on a visual analogue scale (VAS). There was no significant difference in the efficacy between EMLA and placebo creams in pain perception during the entire procedure. There was no significant difference in long term pain perception half an hour after the HSG performance. This study does not support the use of EMLA for HSG.

## Hysterosalpingography: an imaging Atlas with cross-sectional correlation.

*PMID: 25389063 | Indexed in PubMed*

Hysterosalpingography (HSG) provides a unique combination of both fallopian tube and uterine cavity evaluation. A comprehensive understanding of both HSG and correlative cross-sectional imaging findings are essential radiologic skills. This article will review the spectrum of technical artifacts, anatomic variants, congenital uterine anomalies, uterine and tubal pathology, and postsurgical findings as they appear on HSG. Additionally, correlation with MR and ultrasound images is provided. This review article serves as a reference for residents new to HSG as well as staff who perform and interpret HSG infrequently.

## Hysterosalpingography and ultrasonography findings of female genital tuberculosis.

*PMID: 25538038 | Indexed in PubMed*

Genital tuberculosis (TB) is an important cause of female infertility in the world, especially in developing countries. Majority of infertility cases are due to involvement of the fallopian tubes (92%-100%), endometrial cavity (50%), and ovaries (10%-30%); cervical and vulvovaginal TB are uncommon. Genital TB has characteristic radiological appearances based on the stage of the disease process (acute inflammatory or chronic fibrotic) and the organ of involvement. Hysterosalpingography (HSG) and ultrasonography (US) remain the main imaging modalities used in the diagnosis of genital TB. HSG is the primary modality for evaluating uterine, fallopian tube, and peritubal involvement and also helps in evaluating tubal patency. US, on the other hand, allows simultaneous evaluation of ovarian and extrapelvic involvement.

## Contrast induced nephropathy in women with infertility undergoing hysterosalpingography.

*PMID: 39702139 | Indexed in PubMed*

Contrast-induced nephropathy (CIN) defined as an acute kidney injury following the administration of iodinated contrast medium (CM). Hysterosalpingography (HSG) is a radiologic procedure used to investigate the shape and structure of the uterine cavity and the patency of the fallopian tubes in the evaluation of infertility. To date, there have been no reports evaluating the development of CIN after HSG procedure. Therefore, we investigated whether CIN development occurs in infertile women who underwent HSG and its relationship with clinical and laboratory changes in women who underwent HSG. This study was undertaken in 65 women who had infertility evaluation, uterine anomalies and/or tubal blockages. CIN was defined as a 25% relative increase, or a 0.5 mg/dL (44 µmol/L) absolute increase, in serum baseline creatinine (SCr) within 72 h of contrast exposure in the absence of alternative conditions. Hysterosalpingography (HSG) was performed using 5-20 ml of contrast medium. All patients performed routine laboratory tests including assessment of serum creatinine and urea and estimated glomerular filtration rates before and 2-3 day after HSG. Statistical analysis was performed with MedCalc Statistical Software Program v22.023 (Ostend, Belgium) program. The mean ages of participants were 29.5 years and mean BMI were 26.2 kg/m2. The rate of CIN was 12.3% and the severe nephropathy was 1.5% in our study population. The baseline SCr level was $0.59 \pm 0.06$ mg/dL in women with CIN and $0.67 \pm 0.11$ mg/dL in women without CIN. The baseline SCr level was significantly lower in CIN group that non-CIN group ($p = 0.0309$). The SCr level significantly higher in CIN group than non-CIN group 48-72 h after HSG ($p = 0.0005$). In the multivariate logistic regression analysis, the baseline SCr was found an independent risk factor for the prediction of CIN in women who underwent HSG. The HSG procedure is generally a safe method, but the iodine-containing contrast material used in HSG may be associated with temporary adverse effects on kidney function.

## Hysterosalpingography Observations in Female Genital Tuberculosis with Infertility.

*PMID: 37033134 | Indexed in PubMed*

Hysterosalpingography (HSG) is radiographic evaluation of uterine cavity and tubal patency. The aim of this study was to evaluate the safety and utilisation of HSG in female genital tuberculosis (FGTB) with infertility. The study was conducted in a tertiary referral centre of North India. It was a prospective study on 87 cases of FGTB with infertility. Diagnosis of FGTB was made by composite reference standard using the presence of acid-fast bacilli on microscopy/culture or positive GeneXpert, positive polymerase chain reaction or epithelioid granuloma on endometrial biopsy or definitive or probable findings on laparoscopy or hysteroscopy. Suitable statistical methods were used with STATA software version 12.0. HSG findings were normal in 49 (56.32%) cases. There were filling defects in 14 (16.09%), short and shrunken cavity in 4 (4.49%), intrauterine synechiae in 14 (16.09%), T-shaped cavity in 3 (3.44%) and deformed uterine cavity in 5 (5.74%) cases. Fallopian tube findings were hydrosalpinx in 12 (13.79%) and 11 (12.64%) cases, beading of tube in 4 (4.59%) and 2 (2.29%) cases, pipestem appearance in 2 (2.29%) cases each and Maltese cross appearance in 3 (3.44%) and 2 (2.29%) cases, respectively. Tubal blockage was seen in 69 (79.31%) and 67 (77.01%) cases being cornual block in 28 (32.18%) and 26 (29.88%) cases, mid-tubal block in 16 (18.39%) and 15 (17.24%) cases, multiple blocks in 10 (11.49%) and 12 (13.79%) cases and fimbrial block in 15 (17.24%) and 14 (16.09%) cases. None of the cases had flare-up of the disease after HSG in the current study. HSG is a useful modality in FGTB with infertility.